

For questions 1 through 4, Circle A is shown. $\overline{CD}$ and $\overline{DB}$ are tangent to the circle.

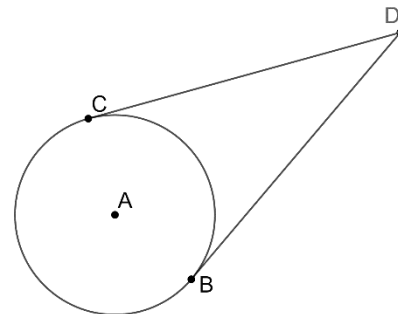
$CD = 10x + 30$ and $DB = 4x + 66$.

1. What is the value of x ?

$$10x + 30 = 4x + 66$$

$$6x = 36$$

$$x = 6$$



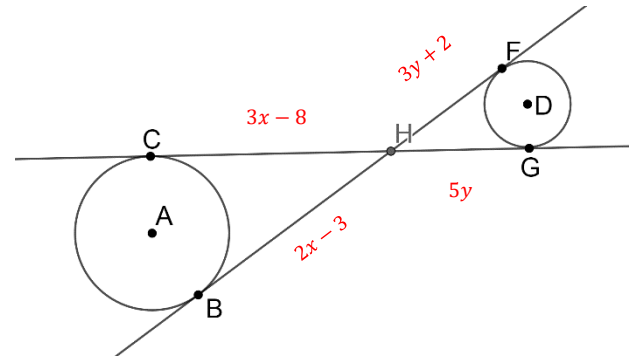
2. What is the length of $\overline{CD}$?

$$CD = 10(6) + 30 = 90$$

3. What is the length of $\overline{DB}$?

$$DB = 4(6) + 66 = 90$$

For questions 4 through 10, Circles A and D are shown. Lines $\overline{CG}$ and $\overline{BF}$ intersect at point H . Line $\overline{CG}$ is tangent to circle A at point C and circle D at point G . Line $\overline{BF}$ is tangent to circle A at point B and circle D at point F . $CH = 3x - 8$, $HG = 5y$, $BH = 2x - 3$, $HF = 3y + 2$.



4. Write an equation you can use to solve for x .

$$3x - 8 = 2x - 3$$

5. Solve the equation you wrote in question 4 for x .

$$3x - 8 = 2x - 3$$

$$x = 5$$

6. Write an equation you can use to solve for y .

$$3y + 2 = 5y$$

7. Solve the equation you wrote in question 6 for y .

$$3y + 2 = 5y$$

$$2 = 2y$$

$$y = 1$$

8. What is the length of $\overline{BF}$?

$$BF = 2(5) - 3 + 3(1) + 2 = 12$$

9. What is the length of $\overline{CH}$?

$$CH = 3(5) - 8 = 7$$

10. What is the length of $\overline{HF}$?

$$HF = 3(1) + 2 = 5$$